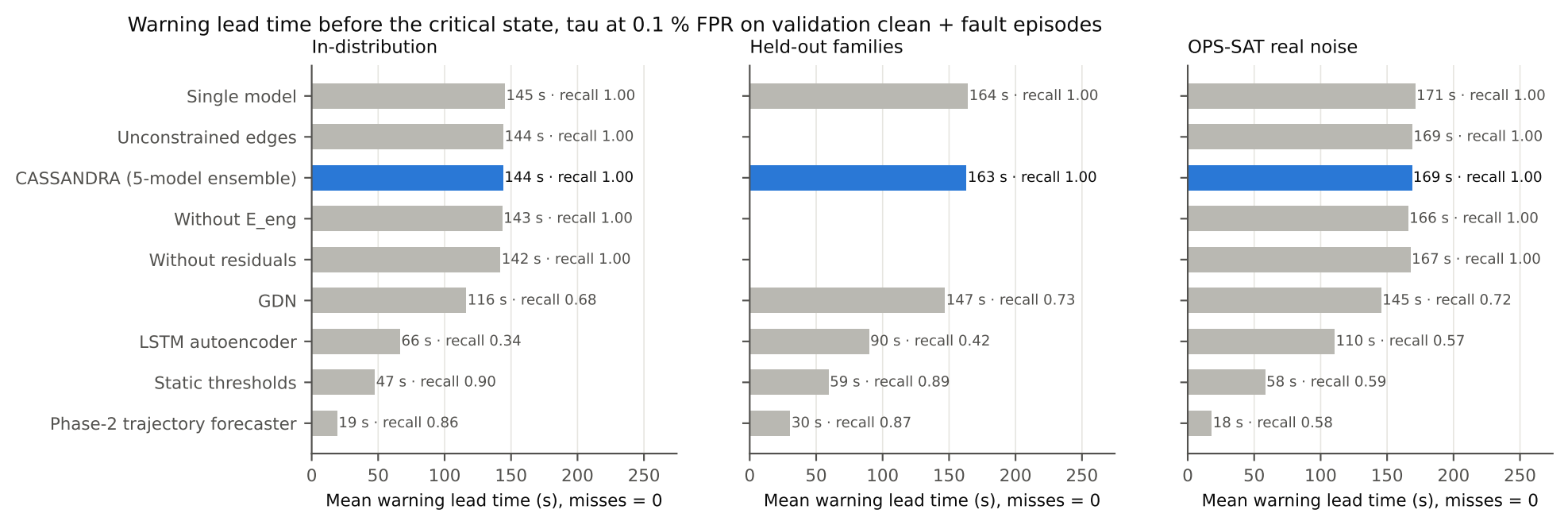
Warning lead time before the critical state, tau at 0.1 % FPR on validation clean + fault episodes
In-distribution
Held-out families
OPS-SAT real noise
Single model
Unconstrained edges
CASSANDRA (5-model ensemble)
Without E_eng
Without residuals
GDN
LSTM autoencoder
Static thresholds
Phase-2 trajectory forecaster
145 s · recall 1.00
144 s · recall 1.00
144 s · recall 1.00
143 s · recall 1.00
142 s · recall 1.00
116 s · recall 0.68
66 s · recall 0.34
47 s · recall 0.90
19 s · recall 0.86
164 s · recall 1.00
163 s · recall 1.00
147 s · recall 0.73
90 s · recall 0.42
59 s · recall 0.89
30 s · recall 0.87
171 s · recall 1.00
169 s · recall 1.00
169 s · recall 1.00
166 s · recall 1.00
167 s · recall 1.00
145 s · recall 0.72
110 s · recall 0.57
58 s · recall 0.59
18 s · recall 0.58
0
50
100
150
200
250
Mean warning lead time (s), misses = 0